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Inside Black Holes

in which physicists, wrestling with Einstein's equation,
seek the secret of what is inside a black hole:
a route into another universe?
a singularity with infinite tidal gravity?
the end of space and time, and birth of quantum foam?

Singularities and Other Universes

What is inside a black hole?

How can we know, and why should we care? No signal can ever emerge from the hole to tell us the answer. No intrepid explorer who might enter the hole to find out can ever come back and tell us, or ever transmit the answer to us. Whatever may be in the hole's core can never reach out and influence our Universe in any way.

Human curiosity is hardly satisfied by such replies. Especially not when we have tools that can tell us the answer: the laws of physics.

John Archibald Wheeler taught us the importance of the quest to understand a black hole's core. In the 1950s he posed "the issue of the final state" of gravitational implosion as a holy grail for theoretical physics, one that might teach us details of the "fiery marriage" of general relativity with quantum mechanics. When J. Robert Oppenheimer insisted that the final state is hidden from view by a horizon, Wheeler resisted (Chapter 6)—not least, I suspect, because of his anguish at losing the possibility to see the fiery marriage in action from outside the horizon.

After accepting the horizon, Wheeler retained his conviction that understanding the hole's core was a holy grail worth pursuing. Just as struggling to understand the evaporation of black holes has helped us to